

Manual

Lab1: Bug Report Classification – ISE Coursework

Overview

This baseline tool automatically classifies a bug report from five deep learning projects Caffe, Incubator-mxnet, Keras, PyTorch and Tensorflow as performance-related (1) or not (0). Using TF-IDF features, it compares three models: Gaussian Naive Bayes (baseline), Linear SVM, and Random Forest.

Setup

1. Clone or download the project repository

```
git clone https://github.com/bmo-bmo/ISE-Bug-Classification.git
```

2. Navigate to the project directory in your terminal

```
cd ISE-Bug-Classification
```

3. Create and activate a virtual environment:

```
python3 -m venv venv  
source venv/bin/activate
```

4. Install required dependencies

```
pip install -r requirements.txt
```

Dataset Setup

The datasets used in this project are five CSV files containing GitHub bug reports from the following deep learning projects: Caffe, Incubator-mxnet, Keras, PyTorch and Tensorflow. For convenience, these CSV files are already included in the data/ folder of the repository. If you need to re-download them, they are sourced from the ISE Lab Solution repository.

If the files need to be redownloaded follow the steps below. The files can be downloaded from: <https://github.com/ideas-labo/ise-lab-solution/tree/main/lab1>

Once downloaded, place all five CSV files inside the data/ folder in the root directory as shown below:

```
ISE_Bug_Classification/  
main.py  
requirements.txt  
data/  
  caffe.csv  
  incubator-mxnet.csv  
  keras.csv  
  pytorch.csv  
  tensorflow.csv
```

The script will automatically locate and process all five files when run.

Running Tool

Ensure the virtual environment is active, then run:

```
python3 main.py
```

The tool will automatically process all five projects in sequence. No additional arguments or configuration are required.

Understanding the Output

For each project, the tool prints a results table showing the average performance of each model across 30 repeated experiments:

Model	Acc	Prec	Rec	F1-Mac	F1-Bin	AUC
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The columns are:

Acc: Accuracy - overall proportion of correct predictions

Prec: Precision (macro) - ability to avoid false positives

Rec: Recall (macro) - ability to find all positive cases

F1-Mac: Macro F1 - harmonic mean of macro precision and recall

F1-Bin: Binary F1 - F1 score for the positive class (performance bugs) only

AUC: Area Under the ROC Curve - overall discrimination ability

Following the results table, statistical analysis is printed showing the Wilcoxon Signed-Rank Test p-value and Cliff's Delta effect size for each proposed model compared against the baseline.